

Perovskite Photovoltaics in July 2026: Buried-Interface Chemistry, Defect Tolerance and the Certified Efficiency Frontier

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Abstract

Perovskite photovoltaics published 648 indexed works in July 2026, of which 181 were read closely and 80 are cited here. This review is organised mechanistically rather than chronologically, and separates what the month's literature verified from what it claimed. The certified frontier advanced on three fronts, each through interfacial rather than bulk chemistry. Noncoplanar multidentate contacts reinforcing the buried interface delivered a 33.1% certified perovskite/silicon tandem [1]; hydrogen-bonding-regulated crystallisation of the wide-bandgap subcell reached 32.13% certified [2]; and homogenisation of the tin-lead distribution through A-site methylammonium cations gave 29.57% certified in an all-perovskite tandem [3]. Single-junction certification clustered near 27.31%, with a soft supermolecule stabilising the buried interface at 27.12% certified and carrying the same chemistry to a 655.2 cm² aperture module [4], the largest certified area in the month. Only 22 of 169 closely read papers reported an independently certified power conversion efficiency at all. Operational-stability evidence was the weakest link: 4 papers reported a time to 80% of initial performance (T80), the longest at 9800 h [5], and 6 named a protocol from the International Summit on Organic Photovoltaic Stability (ISOS), so most lifetime claims this month cannot be compared against each other. Across the corpus an efficiency value was detected in at least 50.4% of abstracts, independent certification in at least 8.3%, and a complete open-circuit voltage, short-circuit current density and fill factor triplet in at least 2.9%. Three mechanistic conclusions follow. Buried-interface chemistry, not absorber composition, now sets the achievable open-circuit voltage. Defect tolerance has overtaken defect elimination as the practical design strategy. Halide homogenisation in wide-bandgap absorbers is the rate-limiting step for tandem integration, and it is the subcell least often optimised in the month's work.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite; tandem solar cells; operational stability

1. Introduction

Perovskite single junctions are now certified above 27% power conversion efficiency (PCE) and perovskite/silicon tandems above 33%, so the field’s open problems have moved from raw efficiency to whether a claimed efficiency is verifiable, whether it survives at module area, and whether it survives operation. High peak efficiencies on small-area cells increasingly diverge from the operational demands of modules, where non-radiative recombination losses at defect sites, interfacial delamination, and volatile species desorption dictate practical viability.

In July 2026, 648 indexed works met the scope gate and 181 were read closely. This review follows four mechanistic threads: buried-interface and contact chemistry, defect tolerance and ion migration, wide-bandgap absorbers for tandems, and operational stability, with scale-up as the constraint coupling them. Interfacial chemical interactions govern carrier extraction and charge accumulation, defect landscapes dictate electric-field-driven ion migration and halide segregation, and wide-bandgap absorbers remain vulnerable to photoinduced phase separation, with all these processes subject to the physical boundaries of scalable deposition.

The month’s headline numbers are strong, the evidence behind them is unevenly reported, and this review separates the two. Critical evaluation distinguishes verifiable physical mechanisms from uncertified champion values, non-standardized aging procedures, and uncharacterized degradation modes.

2. The Certified Frontier in July 2026

Only 22 of 169 closely read papers reported an independently certified power conversion efficiency (PCE), revealing that most literature relies on unverified laboratory metrics. Certified performance ceilings diverge across device architectures. For perovskite/silicon tandems, verified efficiency reached 33.1% in an inverted p-type-intrinsic-n-type (p-i-n) structure [1] alongside 32.13% in two-terminal (2T) devices [2]. In all-perovskite 2T tandems, narrow-bandgap subcell limitations and non-radiative recombination constrained the certified ceiling to 29.57% [3], with lower certified tiers at 28.84% [6] and 27.49% [7]. Single-junction p-i-n perovskite solar cells (PSCs) converged between 27.3% and 27.31% [8,9,10].

Device area exposes a fundamental evidentiary divide: a self-reported champion on a sub-0.1 cm² cell and a certified value on a module represent distinct classes of physical evidence. Uncertified champion micro-cells suppress lateral series resistance and avoid macroscopic defects, allowing self-reported efficiencies to run ahead of certified limits. Conversely, large-area modules must overcome interconnect resistance, transparent conductor sheet resistance, and non-uniform film crystallization. Demonstrating scalable transport, certified p-i-n modules verified 27.31% across both 766 cm² [9] and 684.75 cm² [8].

Beyond these verified benchmarks, uncertified claims overstate large-area performance. Uncertified reports claimed 27.12% on 655.2 cm² [4] and 26.53% on 818 cm² [11], while self-reported in-lab testing on a 625 cm² module claimed 27.8% despite achieving an independent certification of only 27.3% [10]. This discrepancy highlights how uncertified testing masks transient capacitive currents and aperture misalignment, confirming that unverified claims on sub-0.1 cm² cells distort the true technological frontier.

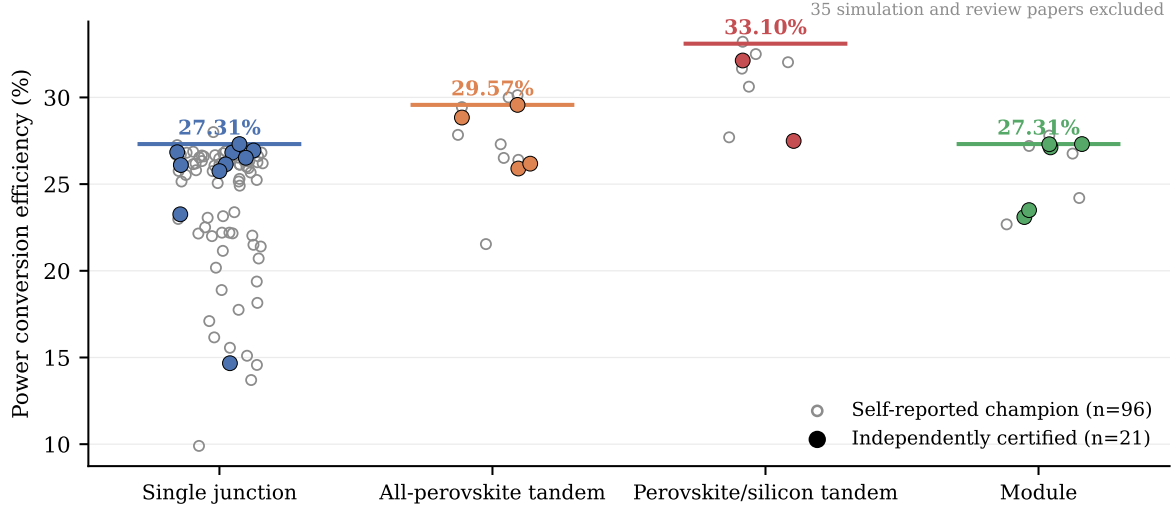


Figure 1. The certified frontier in July 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Buried Interfaces and Contact Chemistry

The buried interface between the hole-transport layer (HTL) and perovskite absorber dictates carrier extraction kinetics and mechanical reliability in inverted perovskite solar cells (PSCs) [12]. Weak chemisorption and disordered self-assembled monolayer (SAM) packing on metal oxides encourage molecular desorption and interfacial delamination; removing loosely bound molecules increases the interfacial fracture energy from 0.94 to 4.82 J m⁻² [13]. To prevent interfacial detachment, multidentate chelating architectures substitute weak monodentate anchoring [8,14,15]. Tetraphosphonic acid iron(III) porphyrins establish tetradentate chelation to oxide substrates, yielding a certified power conversion efficiency (PCE) of 26.82% (27.06% champion) [14]. Bisphosphonate head groups similarly reconstruct nickel oxide (NiOx) interfaces by coordinating undercoordinated lead (Pb²⁺) to achieve a certified 27.31% PCE [8], or reinforce anchoring via hydrogen bonding to retain over 90% of initial performance after 200 h of maximum power point tracking (MPPT) [15]. Tetraphosphorylated phthalocyanines lock interfacial packing to secure a certified 23.1% PCE (22.9% stabilised) on a 20.3 cm² module with zero degradation across 61 days of outdoor testing under International Summit on Organic Photovoltaic Stability (ISOS) ISOS-O-1 protocols [16]. Beyond anchoring strength, self-aggregation via pi-pi stacking creates pinholes that expose bare transparent conductive oxides. Introducing noncoplanar geometry suppresses aggregation to achieve a certified 33.1% tandem PCE (33.3% champion; 23.7% single-junction) [1]. Alternatively, co-adsorbing Lindqvist polyoxometalates increases the SAM anchoring ratio from 29.93% to 53.98% while reducing the desorption rate from 30.7% to 5.6% during continuous MPPT [17]. Interfacial aggregation is also counteracted using ionic liquid lubricants that preserve 92.3% initial efficiency after 800 h at 85 °C under air mass 1.5 global (AM 1.5G) illumination [18], proton-transfer complexes resisting 75 °C degradation for 1368 h [19], or thermally activated in situ polymerization networks that achieve an uncertified 26.87% PCE [20].

At the absorber contact, asymmetric nucleation generates buried microvoids, tensile lattice strain, and deep defect states that induce non-radiative recombination [21,22]. Bidentate phenanthroline molecules chelate Pb²⁺ and facilitate dimethyl sulfoxide (DMSO) solvent evaporation to eliminate

interfacial voids, delivering a 25.15% PCE, but this evidence is constrained to an uncertified 0.04 cm² aperture area [22]. Supramolecular calixarenes mitigate defect propagation via host-guest interactions, yielding a certified 27.12% cell PCE alongside a certified 22.25% on a 655.2 cm² module [4]. Electrostatic alignment and defect passivation proceed concurrently through molecular dipoles: sulfonate-anchored dipotassium salts coordinate undercoordinated tin (Sn⁴⁺) sites on tin oxide (SnO₂) electron-transport layers (ETLs) to passivate oxygen vacancies [23], while fluorinated carbazole self-assembled monolayers (SAMs) passivate halide vacancies, scaling to 1 cm² with 25.06% PCE and 3.11% relative loss [24]. Polycyclic aromatic SAMs with tailored electrostatic potentials (26.20% PCE) [25], multifunctional modifiers (26.67% PCE) [26], ternary phosphine bridges stabilizing formamidinium cations [27], and preburied two-dimensional (2D) perovskite inducers that invert tensile to compressive strain [21] similarly regulate the heterojunction. Nevertheless, theoretical one-dimensional Solar Cell Capacitance Simulator (SCAPS-1D) simulations predict ideal efficiencies exceeding 30%, including 30.81% at 300 K [28] and 29.1% in HTL-free bilayer ETLs [29], yet these drift-diffusion calculations omit the spatial heterogeneity of SAM desorption, localized pinholes, and void formation that dominate physical devices.

4. Defect Tolerance, Passivation and Ion Migration

Non-radiative recombination at carrier-selective contacts in perovskite solar cells (PSCs) is driven by undercoordinated lead(II) (Pb²⁺) cations, dangling bonds, and halide vacancies that require Lewis acid-base coordination [30,31,32,33,34]. At hole-transport layer (HTL) contacts, carbazole-based self-assembled monolayers (SAMs) coordinate Pb²⁺ to reach 26.85% power conversion efficiency (PCE) [30], while carboxyl-sulfur molecules coordinate dangling bonds and free lead(II) iodide (PbI₂) for an uncertified 26.7% PCE [32]. At electron-transport layer (ETL) contacts, 1-(4'-sulfophenyl)-3-carboxyl-5-pyrazolone sodium salt (SCP) yields a certified 26.84% PCE (champion 27.25%, stabilised 26.84%) with 90% retention after 1485 h of maximum power point tracking (MPPT) [31]. Amphoteric ionic liquids passivate both acidic and basic sites for 26.13% champion PCE (20.06% PCE and 85.74% fill factor (FF) in 1.84 eV cells) [33], while Pb-N coordination and vacancy filling preserves 88.67% efficiency after 1000 h MPPT at 65 °C (uncertified 25.06% PCE) [34]. Passivating tin(IV) oxide (SnO₂) oxygen vacancies with conjugated sorbate anions yields an uncertified 22.16% PCE (open-circuit voltage (Voc) 1.183 V, FF 84.30%) [35]. Plasma surface engineering passivates laser scribes, securing a certified 23.5% PCE on a 65.05 cm² module and 27.2% small-cell champion PCE (98.1% retained after 2000 h MPPT at 85 °C) [36].

Electric-field- and heat-induced migration of halides and A-site cations accelerates phase segregation, requiring chemical trapping and dimensional barriers [37,38,39,40]. Dual iodide-trapping molecules immobilize mobile halides at the HTL interface for 26.81% champion PCE and a T93 lifetime over 1140 h at 85 °C [37], while acetylthiocholine chloride (ATC) coordinates Pb²⁺ and fills vacancies to suppress segregation, yielding 22.68% champion PCE on a 63.7 cm² module [40]. Two-dimensional/three-dimensional (2D/3D) heterojunctions formed by halide imidazolium cations block ion pathways, retaining 92% efficiency after 1200 h at 85 °C (26.33% champion PCE) [39]. In situ two-dimensional (2D) capping via hydrogen bonds with formamidinium (FA⁺) retains 90% efficiency after 800 h at 85 °C in N₂, yet evaluation on a 0.072 cm² aperture (26.32% champion PCE) leaves scalability unproven [41]. Molten salt dissolution of 2D phases suppresses random nucleation and eliminates interfacial PbI₂, boosting PCE from 23.43% to 25.73% [42]. Unified ligand-dimensional designs halt cation migration, achieving certified 26.52% PCE (26.88% champion) on small cells and certified 17.75% PCE (20.25% uncertified) on 900 cm² modules, retaining over 91% efficiency after

1000 h at 85 °C and 85% relative humidity (RH) [38].

Residual tensile strain accelerates defect nucleation and ion transport, coupling mechanical stress to electronic loss [43,44,45,46,47]. Coordinating adjacent Pb²⁺ with dithiopyr relieves lattice strain and directs preferential (100) orientation, yielding certified 26.14% PCE (26.9% champion) and 21.28% on 30 × 30 cm² modules [43]. Cross-linked 2-amino-3-hydroxybutanoic acid (AHBA) networks relax buried SnO₂ interface strain and accelerate electron extraction to 3.5 ps for 26.32% champion PCE [44]. Low-temperature liquid-assisted vibrational stress-relief (LTL-VSR) relieves macroscopic stress without additives, yielding 26.47% rigid and 24.64% flexible cell efficiencies, alongside 23.59% (16.8 cm²) and 20.55% (600 cm²) module efficiencies [46]. Photoresponsive fullerene C₆₀-azo generates compressive stress under ultraviolet (UV) light via trans-to-cis isomerization, retaining 92.7% performance after 488 h UV exposure with 26.6% champion PCE [45]. Finally, phenethylammonium iodide (PEAI) paired with cerium oxide (CeOx) provides redox-buffering passivation, retaining 95% efficiency after 100 thermal shock cycles between 80 °C and -80 °C (at 16 °C/min) with 25.3% champion PCE in poly(triarylamine) (PTAA)-based devices [47].

5. Wide-Bandgap Absorbers and Tandem Integration

Monolithic tandems can surpass the ~29.4% crystalline silicon (c-Si) efficiency limit [48], yet two-terminal (2T) integration remains constrained by halide phase segregation in wide-bandgap (WBG) top absorbers and interfacial non-radiative recombination. In-operando thermal cycling between -100 and +100 °C under air mass 0 (AM0) illumination reveals that WBG subcells segregate rapidly above 50 °C with only partial recovery during dark cooling, accelerating non-radiative voltage loss [49]. Suppressing this demixing requires reinforcing octahedral framework integrity and passivating grain boundaries [50,51,52]. B-site strontium (Sr²⁺) alloying strengthens metal-halide bonding and weakens electron-phonon coupling, yielding uncertified 31.66% power conversion efficiency (PCE) in fully-textured perovskite/silicon tandems with an operational T80 lifetime (time to 80% initial efficiency) of 6,785 hours [50]. At grain boundaries, TPB-Br molecular extrusion drives halogen homogenization to deliver 23.63% PCE in 1.68 electronvolt (eV) subcells and an uncertified 32.03% on 1.05 cm² tandems [51], while conductive ionic-liquid/polyurethane polymer bridges passivate defect states to yield uncertified 20.85% WBG and 25.92% flexible tandem PCEs [52]. Ionic-liquid additives similarly confer radiation resilience, yielding certified 27.49% monolithic tandems over 12.56 cm² that maintain 93% efficiency under proton irradiation [7].

Defective buried contacts and unguided nucleation introduce large open-circuit voltage (V_{oc}) deficits. Retarding nucleation via coordinating solvents directs micrometer-scale grain growth [53,54]: gamma-valerolactone (GVL) coordination with formamidinium (FA⁺) cations achieves uncertified 32.5% champion tandem PCEs [53], while GVL-dimethyl sulfoxide (DMSO) inks yield printed cells exceeding 17% and 28% tandems [54]. Regulating crystallization and passivating defects at hole-transport layers (HTLs) further limits interface recombination [2,55,56]. Inserting a 3-TMA interlayer on self-assembled monolayers (SAMs) regulates crystallization through hydrogen bonding, achieving 23.17% (1.67 eV) and 19.61% (1.84 eV) subcells and a certified 32.13% (33.21% champion) silicon tandem [2]. Bilayer graphene oxide/self-assembled monolayer (SAM) HTLs similarly stabilize triple-junction devices at 27.0% stabilized PCE (27.3% champion) [55], whereas omitting nickel oxide (NiOx) HTLs via hybrid deposition yields flexible perovskite/copper indium gallium selenide (CIGS) tandems with certified 25.9% PCE (26.4% champion, 1.85 V V_{oc}) [56].

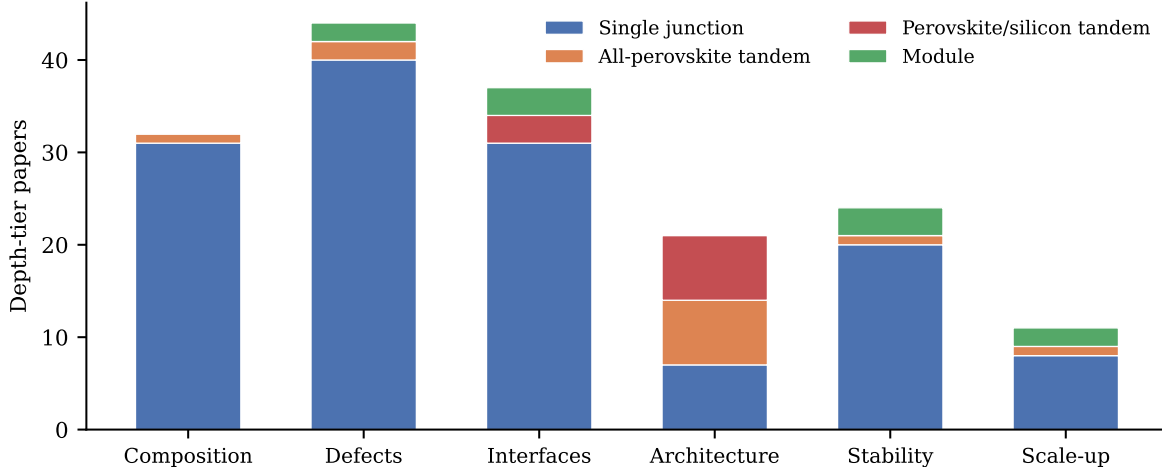


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

6. Operational Stability and Degradation Pathways

Depth-resolved profiling reveals that operational photo-degradation initiates at both buried and top interfaces before propagating into bulk perovskite absorbers [57]. At buried hole-transport layer (HTL) contacts in inverted perovskite solar cells (PSCs), molecular disorder accelerates non-radiative recombination. Rigidifying molecular orientation by coassembling carbazole self-assembled monolayers (SAMs) suppresses interfacial defects, yielding a certified power conversion efficiency (PCE) of 27.31% for single cells and 21.54% for a 766 cm² module under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-1 [9]. Tailoring asymmetric benzothienoindeole SAMs with electron-withdrawing substituents similarly improves packing and energetics, delivering 25.38% on 1 cm² and 20.59% on a 655.2 cm² module [58]. At the top contact, multidentate fullerenes anchor phenethylammonium bromide capping layers against thermal loss, retaining 98.7% initial PCE after 2100 h of maximum power point tracking (MPPT) under continuous illumination and 95.8% after 1800 h at 85 °C under ISOS-L-3 [59], while phase-pure two-dimensional (2D) perovskite reconstruction passivates surfaces [60,61]. Extrinsic ion migration also accelerates failure: lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI) dopants in HTLs release mobile lithium ions that drift under bias, whereas volatile dopants eliminate ionic residues [62]. Surface passivation via aromatic carbonyl molecules yields 26.12% PCE, but only on an uncertified, sub-0.1 cm² area (0.09 cm²) [63].

In wide-bandgap (WBG) mixed-halide absorbers, operational instability stems from electric-field-induced halide segregation driven by trapped-charge accumulation [64]. Operating cells under maximum stability point tracking (MSPT) instead of MPPT alleviates trapped carriers, retaining approximately 90% initial power after 700 h [64]. Chemical stabilization targets the same thermodynamic vulnerability: incorporating 1,4-phenylenediamine dihydriodide into 1.93-eV absorbers raises the ion migration activation barrier to suppress phase segregation, achieving a certified PCE of 26.18% with an open-circuit voltage (Voc) of 3.148 V on 1.17 cm² triple-junction cells [65]. Rubidium incorporation similarly homogenizes vertical cesium distribution in 1.67-eV pure-iodide perovskites to prevent demixing, reaching an uncertified 22.51% PCE under ISOS-L-1 [66].

Under thermal and light fluctuations, dynamic lattice strain drives degradation [67]: residual tensile

stress relaxes via diffusion-mediated inelastic deformation, accelerating moisture uptake by an order of magnitude [68]. Interfacial networks and printed silicon oxynitride barrier films counter moisture penetration, the latter retaining $93\% \pm 5\%$ PCE after 1000 h at 25 °C and 60% relative humidity (RH) under ISOS-D-1 [69,70,71]. Yet a methodological gap obscures durability: only 4 of 169 closely read studies reported the time to 80% initial efficiency (T80), spanning 200 h, 720 h, 6785 h, and 9800 h [5,50,72,73], and only 6 specified an ISOS protocol [9,11,16,17,59,66]. Prevailing tests apply constant 1-sun illumination at mild temperatures, probing photocarrier collection while failing to evaluate thermomechanical fatigue, reverse bias, or dark-light cycles.

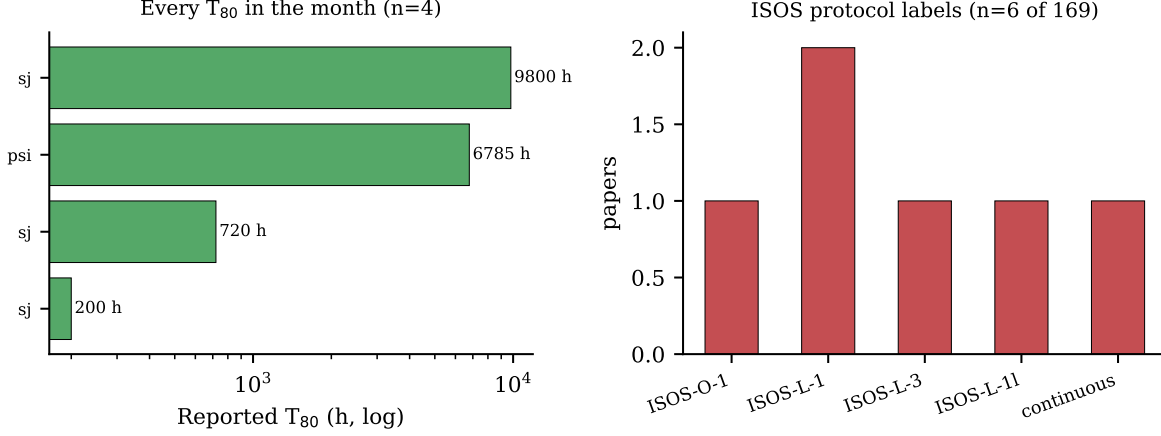


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

7. Scale-Up and the Area Penalty

The area penalty during module scale-up stems from three coupled physical bottlenecks: series-resistance losses across monolithic interconnects, non-uniform crystallization in scalable coating, and localized interfacial recombination [10,74,75]. Extrinsic integration and resistive interconnect losses rival intrinsic material recombination in driving the efficiency drop from discrete cells to series-connected modules [74,75]. Hyperspectral luminescence imaging attributes this deficit to resistive transport across interconnections in 64 cm² modules [74], while electrostatic templating with indol-3-ylethylammonium iodide spacer cations leaves a certified power conversion efficiency (PCE) gap between 26.94% on small-area cells and 23.21% on a 25 cm² monolithic module [75]. Aperture scaling demonstrates that modules can maintain high efficiencies: a certified 27.3% PCE (champion 27.8%) across 625 cm² with nonfullerene electron-transport layers (ETLs) [10], alongside uncertified champion outputs of 27.12% on 655.2 cm² [4], 27.31% on 684.75 cm² [8], 27.31% on 766 cm² [9], and 26.53% on 818 cm² [11].

Transitioning from spin-coating to scalable meniscus blade coating or vapor methods induces drying gradients and phase degradation [76,77,78,79]. Incorporating Lewis base and zwitterion dual additives into blade-coated tin-lead perovskite solar cells (PSCs) yields grains over 800 nm and elevates open-circuit voltage (Voc) from 0.797 to 0.850 V, achieving 21.02% single-junction PCE and a two-terminal (2T) tandem champion PCE of 27.84% (27.33% stabilised) [76]. For ambient fabrication, surface-confined protection stabilizes pre-annealing crystallization, enabling air-processed blade-coated methylammonium (MA)-free PSCs to achieve 26.14% champion and 25.75% certified

PCE, with 93.11% retention after 1300 h of continuous 1-sun maximum power point tracking (MPPT) [77]. Vapor-phase annealing (VPA) establishes uniform thermal-humidity fields to direct bottom-up crystallization, yielding an additive-free champion PCE of 22.15% [78], whereas stabilizing precursor solutions against aging via anion- π interactions preserves 25.76% efficiency after 7 days and 23.88% after 3 months [79]. High-throughput quantification of microstructural homogeneity is assisted by deep segmentation frameworks such as PerovSegNet, which reaches 87.25% mean average precision (mAP) in scanning electron microscopy (SEM) image analysis at 25.22% lower computational cost [80]. Replacing fullerene ETLs with the nonfullerene conjugated polymer 2PB-T also addresses contact degradation, retaining over 98.6% of initial PCE after 1752 h of continuous MPPT at 85 °C in air on small-area cells [10].

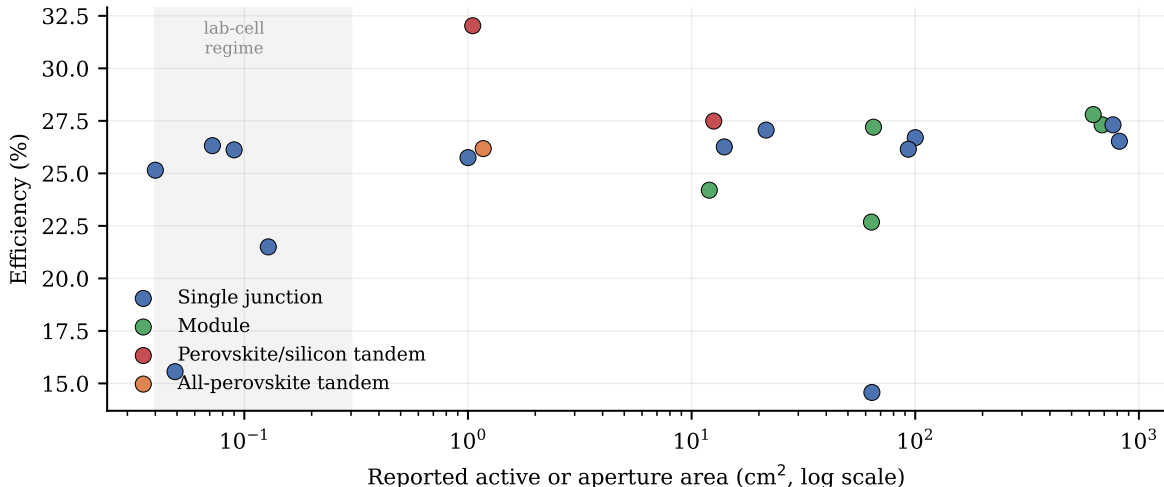


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

8. Research Gaps and Outlook

The performance divergence between single junctions and tandems defines the first gap: the best certified perovskite/silicon tandem this month was 33.1% power conversion efficiency (PCE) [1], while certified single junctions clustered near 27%, showing that the wide-bandgap (WBG) sub-cell limiting the tandem is not the one most papers optimise. Chemical treatments claimed to homogenize mixed halides in pristine WBG absorbers leave unresolved whether halide homogenisation suppresses phase segregation under operating bias, as carrier gradients trigger iodide clustering [49,64]. Operando nanoscale photoluminescence spectroscopy during continuous maximum power point tracking (MPPT) load under elevated operating temperatures would establish whether compositional uniformity survives operational electric fields.

A second gap concerns buried-interface chemistry that survives thermal cycling rather than only lifting initial efficiency. Molecules and self-assembled monolayers (SAMs) passivate buried defects to lift open-circuit voltage (V_{oc}) on pristine substrates, but thermal expansion mismatches induce cyclic shear stresses that cause delamination [1,19]. In situ grazing-incidence wide-angle X-ray scattering with operando transient photovoltage measurements during temperature cycling across thermal extremes would quantify strain relaxation and resolve whether interfacial bonding withstands thermal fatigue.

A third gap lies in ion-migration timescales measured under load rather than in the dark. Only 4 of 169 closely read papers reported a time to 80% of initial performance (T80) lifetime, the longest 9800 h [5], short against a 25-year module warranty. Extracting kinetics from dark impedance ignores how photogenerated carriers screen fields and alter vacancy formation enthalpies under illumination [64,65]. Intensity-modulated photovoltage spectroscopy under bias illumination would capture true ionic drift velocities and accumulation rates under load.

The fourth gap is whether passivation that lifts Voc on a 0.1 cm² cell survives slot-die coating at module area. Only 27 of 648 works addressed scale-up at all, and the largest reported aperture was 818 cm² [11], orders of magnitude below a commercial module. Meniscus coating promotes passivator aggregation, forming pinholes across large areas [75]. Operando photoluminescence imaging synchronized with optical reflectometry during slot-die deposition would track drying kinetics and isolate spatial Voc losses across modules. Next month’s literature would gain immediate utility if studies paired small-cell efficiencies with operando stability under load and reported spatial coating uniformity.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

This manuscript was prepared as part of our Research Project for making a reliable Autonomous Researcher Agent. The agent executed the monthly review pipeline without human intervention: it harvested the indexed literature for the month through the OpenAlex, Semantic Scholar, Crossref and arXiv interfaces, applied a deterministic topical filter, ranked and selected the papers read closely, extracted every quantitative claim into a structured record binding each number to a verbatim quotation from that paper’s own abstract, computed the statistics and figures from those records, and drafted this text from the resulting evidence briefs. Literature harvesting, selection, statistics, figures and gate checking are deterministic scripts rather than model output. Structured extraction used qwen3.8-flash through the opencode interface; the prose was written by gemini-3.8-flash-high through the agy interface; independent review used opus at high reasoning effort. No figure is a model-generated image: every panel is plotted directly from the extracted records. Every quantitative statement in the main text resolves to the machine-generated statistics file shipped as Supplementary Information, and every citation resolves to a record carrying its source quotation.

The author reviewed and edited the content, verified the selection, and takes full responsibility for the content of the published article.

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